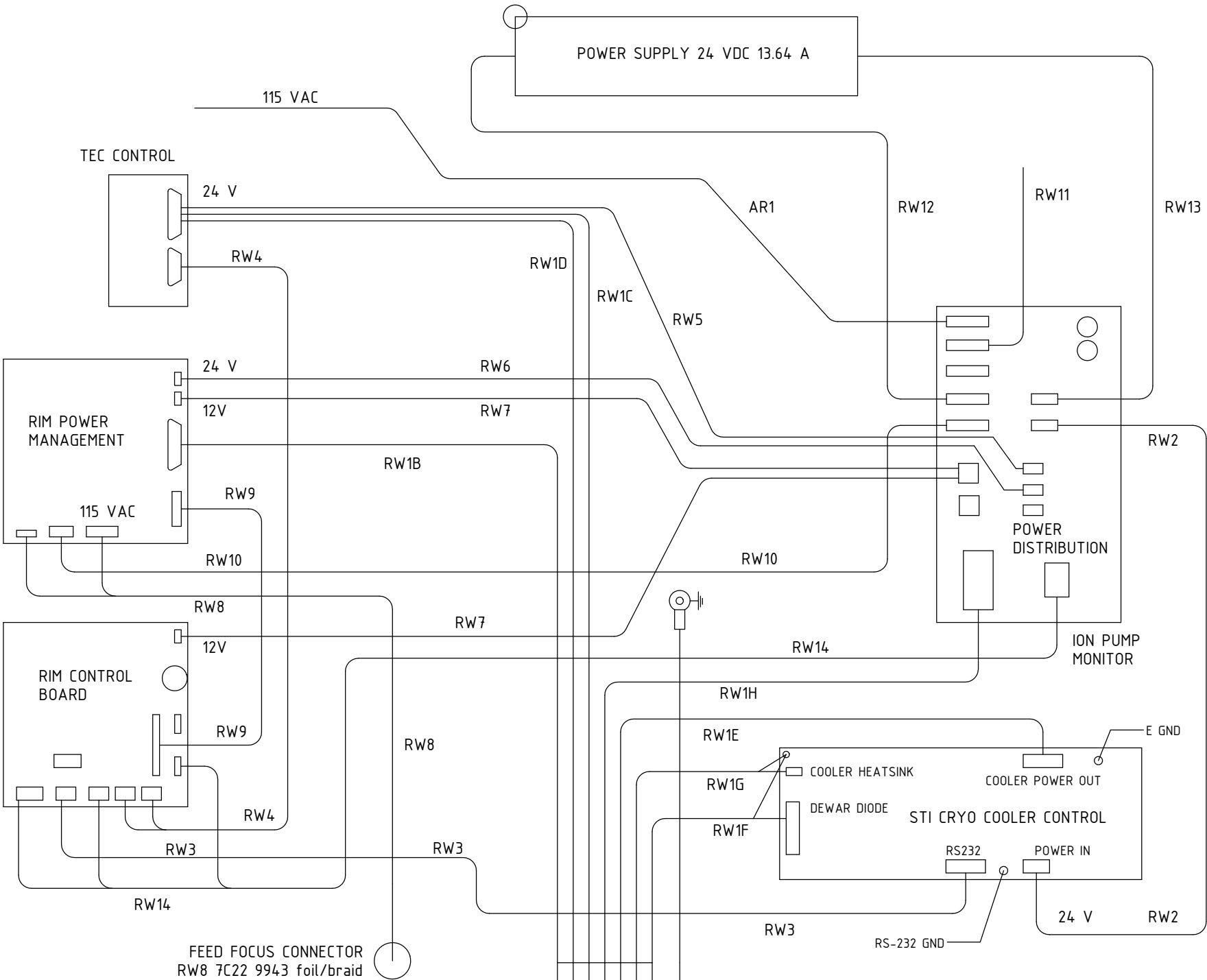


D

C

B

A



2007-05-09 RW1H PRELIMINARY ION PUMP CONTROL CABLE
Belden 9943 or 9611 x 96" LONG WITH molax 43025-0800 AND Amphenol 7 pin

IDENTIFIER	DESCRIPTION
RW1B	PAX
RW1C	TEC SENSE
RW1D	TEC POWER
RW1H	ION PUMP
RW1E	CRYO POWER
RW1G	CRYO TEMP
RW1F	DEWAR DIODE
RW1J	GROUND MAIN

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NAME	DATE
DRAWN M. C. FLEMING	2008-02-11
CHECKED	
MATERIAL	
FINISH	

UNLESS OTHERWISE SPECIFIED DIMENSIONS ARE INCHES & DEGREES		
TOLERANCES:	LINEAR	ANGLE
WHOLE & FRACTION	± 1/32	± 1°
DECIMAL X.X	± 0.1	± 0.1°
DECIMAL X.XX	± 0.01	± 0.01°
DECIMAL X.XXX	± 0.005	± 0.001°
BREAK SHARP EDGES 0.010 RAD ± 0.003 MACHINED SURF FINISH < 125 MICROINCH		
WEIGHT:		
SCALE:		

Minex Engineering Corp.

1000 Apollo Ct., Unit G, Antioch, CA 94509
Tel: 925-757-6785 Fax: 925-757-1083

**Rim Box
Wiring Diagram**

B 30-27-502 1 of 7 B

SIZE DWG NUM: SHEET OF REV.

REVISIONS

REV	DESCRIPTION	ZONE	NAME	DATE
A	RELEASED		M.C.F.	2002-00-00
B	NEW RW1 CABLE TO PAX CASE		M.C.F.	2008-05-03

NOTES:

Diagram = Drawing showing wire and cable routing between major components. It usually gives some indication of the physical location of components relative to each other

Schedule = Spreadsheet showing wire and connector types, pin assignments, color codes, lengths etc.

Schematic = Drawing showing logical electrical connections between components, values of components and component lists, usually for boards.

30-27-500 is a Engineering Part Secification document number.
30 is for ATA project
24 is for Feed
26 is for enclosures and conduits.
27 is for air, electrical, or fiber
500 is just a sequential number system.

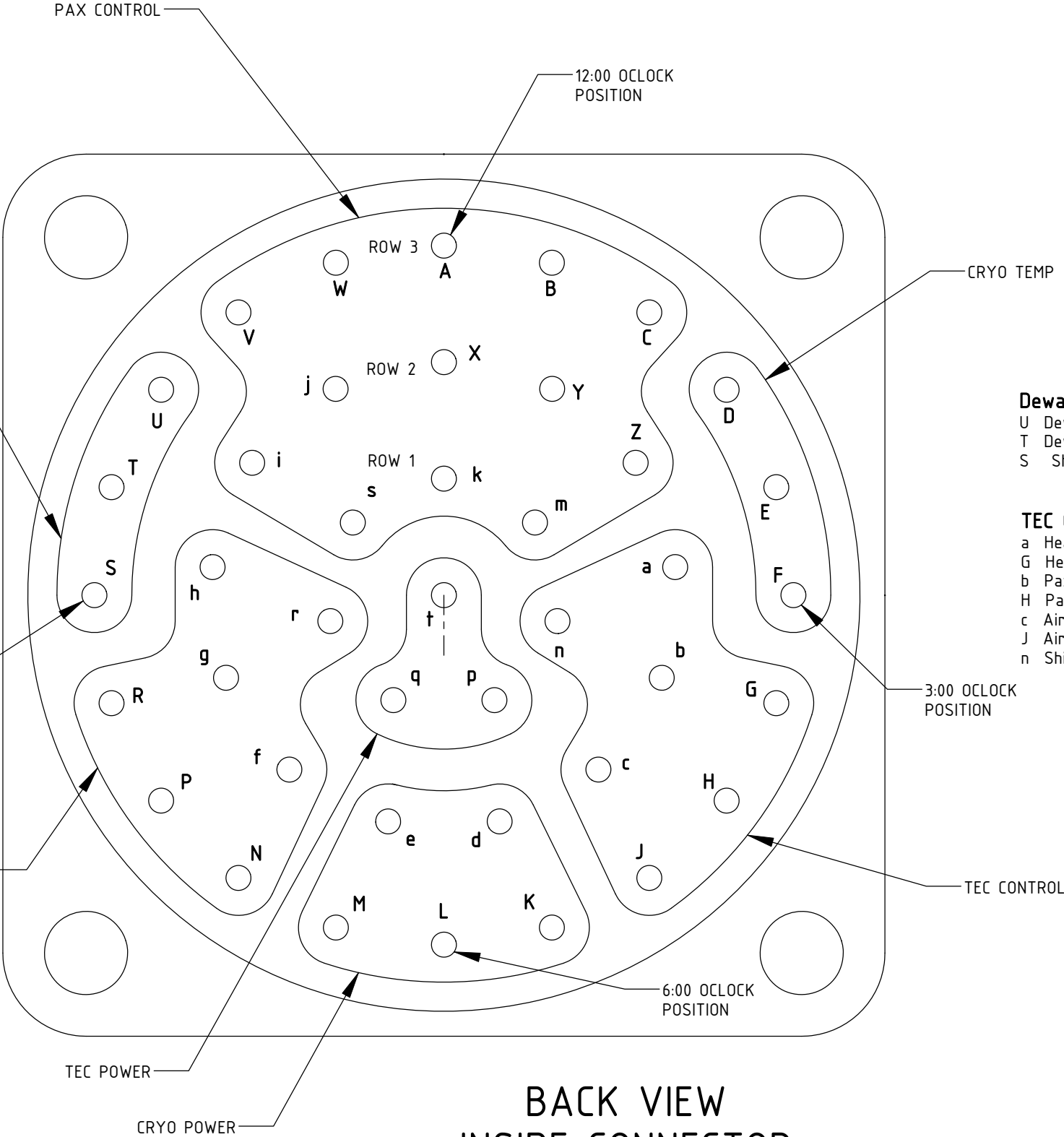
REFERENCE THE FOLLOWING DOCUMENTS

30-27-100 Antenna wiring Documents
30-27-200 Alidade wiring Documents
30-27-203 Alidade wiring Schedule (excel file)
30-27-500 Rim Wiring Documents (general)
30-27-502 Rim Wirng Diagram (drawings)(this drawing)
30-27-503 Rim wiring Schedule (excel file)
30-27-520 Rim Power & Ion Monitor Board
30-27-530 Rim Control Board
30-27-540 Rim Power Management Board
30-27-550 TEC Control Board
30-27-560 STI Cryo Control Board

30-24-300 PAX Case Assy. (mechanical)
30-24-360 Ion Pump Control. (inside PAX case)
30-26-370 Rim Power & Ion Monitor (mechanical)
30-26-230 Cable Assy Rim Box to PAX Case
30-26-234 Connector Amphenol AL00F21-41S

REVISIONS				
REV	DESCRIPTION	ZONE	NAME	DATE
A	RELEASED			

DOCUMENT 30-27-503 RIM WIRING SCHEDULE
SUPERCEEDS THE INFORMATION SHOWN HERE



PAX Control 13 pins

- Z Gnd OTXTEC
- Y +5V OTXTEC
- B Gnd OTX laser / PAM
- C +6V OTX laser / PAM
- k Gnd Sense
- s +6 Sense
- X +5 Sense
- m -6 Sense
- j +5V OTXTEC
- i Gnd OTXTEC
- W -6V OTX / PAM
- V Gnd OTX / PAM
- A Shield (not used ?)

Cryo Heatsink Temp 3 pins

- D Cryo Temp 1
- E Cryo Temp 2
- F (not used)

Dewar Temp 3 pins

- U Dewar Temp 1
- T Dewar Temp 2
- S Shield (not used ?)

Ion Pump Control 7 pins

- f Current monitor
- g External reset
- h Power +12 VDC
- r Power com
- N Shield (not used ?)
- P Fault high voltate
- R Fault blown fuse

TEC Power 3 pins

- p TEC Power
- q TEC Power
- t (not used)

Cryo Power 5 pins

- d Cryo Power 1a
- K Cryo Power 1b
- e Cryo Power 2a
- M Cryo Power 2b
- L Shield (not used ?)

TEC Control 7 pins

- a Heat Sink Temp 1
- G Heat Sink Temp 2
- b Pax Temp 1
- H Pax Temp 2
- c Airstream Temp 1
- J Airstream Temp 2
- n Shield (not used ?)

Shielding issues have not been finalized.
Some cables may have shields grounded at the rim box end
and other cables may be grounded atPAX case
or at desination inside the feed.
We have not determined which shields must pass through the connector.

Extreme care must be taken to prevent cross talk onto Dewar Temp cable.
The Cryo Power cable is most likely the worst emmitter of noise.
In tests 2008-02-08 The Cryo Power (Belden 83322E no foil & 90% braid)
will foul the Dewar Temp (Alpha 3211 no foil 90% braid) with only 3" of
parallel contact. The Dewar Temp cable will be changed in the future.

**BACK VIEW
INSIDE CONNECTOR
RW1-A**

All connecors viewed from the back. That would be
the view as contacts are inserted form the back,
or the view would match a view into the mating part.

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NAME	DATE
DRAWN	
CHECKED	
MATERIAL	
FINISH	

UNLESS OTHERWISE SPECIFIED DIMENSIONS ARE INCHES & DEGREES		
TOLERANCES:	LINEAR	ANGLE
WHOLE & FRACTION	±	±
DECIMAL X.X	±	±
DECIMAL X.XX	±	±
DECIMAL X.XXX	±	±
BREAK SHARP EDGES 0.010 RAD ± 0.003 MACHINED SURF FINISH < 125 MICROINCH		
WEIGHT:		
SCALE:		

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**Rim Box
Wiring Diagram**

B	30-27-502	2 of 7	B
SIZE	DWG NUM:	SHEET	OF
			REV.

8

7

6

5

4

3

2

1

REVISIONS				
REV	DESCRIPTION	ZONE	NAME	DATE
A	RELEASED			

All connectors viewed from the back. That would be the view as contacts are inserted from the back, or the view would match a view into the mating part.

RW1B
PAX Control

RW1C & RW1D + (RW5)
TEC Power & Sense

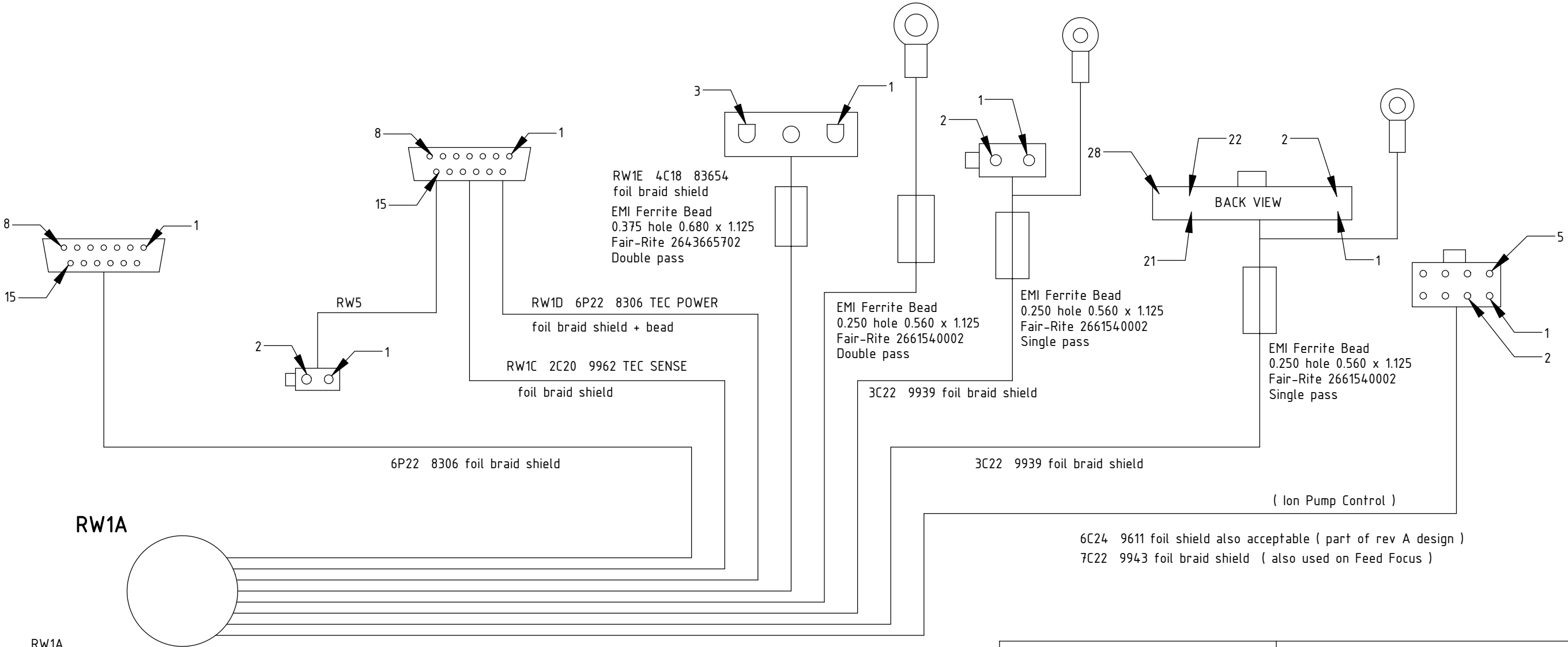
RW1E
Cryo Power

RW1J
Main Ground

RW1G
Cryo Heatsink Temp

RW1F
Dewar Temp

RW1H
Ion Pump Control



RW1A
Amphenol
Series 38899 40 sockets
See Sheet 2

On Tree
Annotations (right click)
Display Reference Dimensions

DOCUMENT 30-27-502 RIM WIRING SCHEDULE
SUPERCEDES THE INFORMATION SHOWN HERE

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NAME	DATE
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CHECKED	
MATERIAL	
FINISH	

UNLESS OTHERWISE SPECIFIED DIMENSIONS ARE INCHES & DEGREES		
TOLERANCES:	LINEAR	ANGLE
WHOLE & FRACTION	±	±
DECIMAL X.X	±	±
DECIMAL X.XX	±	±
DECIMAL X.XXX	±	±
BREAK SHARP EDGES 0.010 RAD ± 0.003 MACHINED SURF FINISH < 125 MICROINCH		
WEIGHT:		
SCALE:		

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**Rim Box
Wiring Diagram**

B	30-27-502	3 of 7	B
SIZE	DWG NUM:	SHEET	OF
			REV.

8

7

6

5

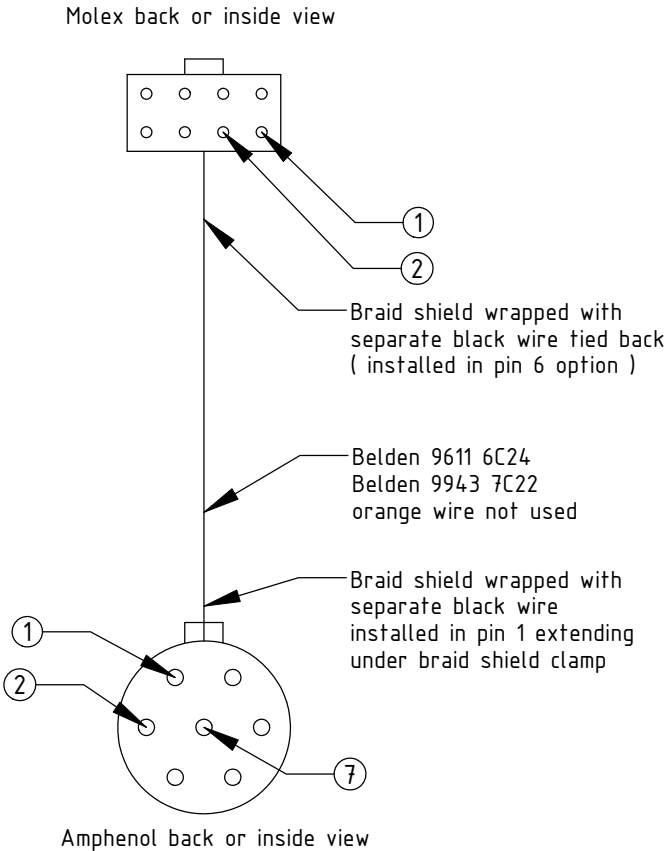
4

3

2

1

REVISIONS				
REV	DESCRIPTION	ZONE	NAME	DATE
A	RELEASED			



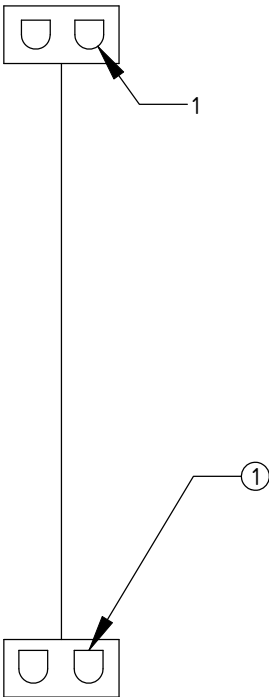
**Special Test Cable
similar to RW1H**

Molex 43025-0800/43030-0008

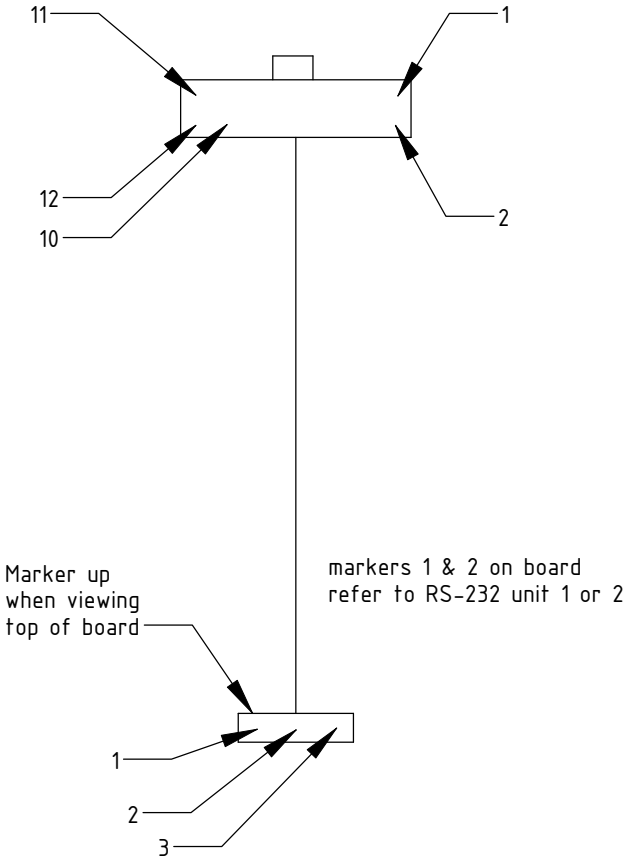
- | | | |
|---|-------|-----------------------|
| 1 | Blue | Current monitor |
| 2 | N | |
| 3 | White | External reset |
| 4 | Red | Power +12 VDC |
| 5 | Black | Power com |
| 6 | Black | Shield (wire added) |
| 7 | Brown | Fault high voltage |
| 8 | Green | Fault blown fuse |

Amphenol C 091

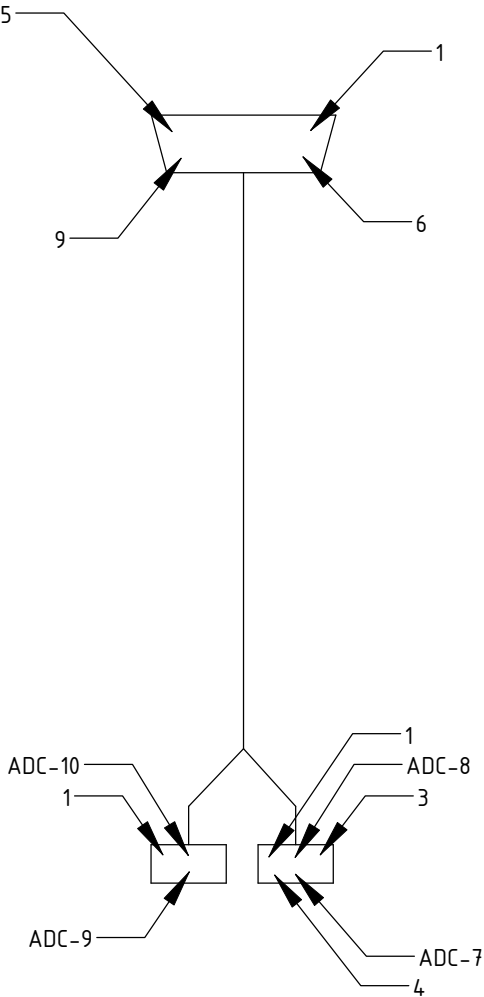
- | | | |
|---|--------|-----------------------|
| 1 | Black | Shield (wire added) |
| 2 | Black | Power Common |
| 3 | Red | Power +12 VDC |
| 4 | Orange | Fault blown fuse |
| 5 | Brown | Fault high voltage |
| 6 | White | External reset |
| 7 | Blue | Current monitor |



RW2



RW3



RW4

Molex SL and Molex Micro Fit position number convention does not match. When viewed from the back, SL uses pos 1 Upper Right (UR), whereas Micro-Fit uses pos 1 Lower Left (LL).

All connectors viewed from the back. That would be the view as contacts are inserted from the back, or the view would match a view into the mating part.

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MATERIAL	
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TOLERANCES:	LINEAR	ANGLE
WHOLE & FRACTION	±	±
DECIMAL X.X	±	±
DECIMAL X.XX	±	±
DECIMAL X.XXX	±	±
BREAK SHARP EDGES 0.010 RAD ± 0.003 MACHINED SURF FINISH < 125 MICROINCH		
WEIGHT:		
SCALE:		

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Rim Box Wiring Diagram				
B	30-27-502	4 of 7	B	
SIZE	DWG NUM:	SHEET	OF	REV.

